

HW02 Main Report - Domain Testing and Boundary Value Analysis on EShop

1. Introduction

This report consolidates the full Domain Testing and Boundary Value Analysis content from sections 1 to 7 of each selected feature file in `HW02/test-cases/`, with the assumption-verification subsections omitted as requested.

Included source files:

- `HW02/test-cases/FR05-product-listing-search.md`
- `HW02/test-cases/FR09-discount-coupons.md`
- `HW02/test-cases/FR17-coupon-management-crud.md`
- `HW02/test-cases/Mobile-FR04-profile-management.md`

2. Overall Test Summary

Feature	Designed	Executed	Passed	Failed	Not Executed	Bugs Found
FR-05 Product Listing and Search	15	12	12	0	3	0
FR-09 Discount Coupons	23	23	12	7	0	3
FR-17 Coupon Management CRUD	54	54	24	25	0	8
Mobile-FR04 Profile Management	39	39	30	9	0	9
Total	131	128	78	41	3	20

Notes:

- Bugs Found follows each feature test-case execution table and consolidated bug report.
- Confirmed and potential defect groups are documented in `HW02/reports/bug-report.md`.
- Requirement-unclear and inconclusive cases are treated as executed when the tester recorded an actual result, but they are not counted as passed.

3. FR-05 Product Listing and Search

Source: `HW02/test-cases/FR05-product-listing-search.md`

1. Feature Understanding

Item	Details
Feature ID	FR-05
Feature name	Product Listing and Search
User role	Visitor or logged-in customer
Main purpose	Display all products on the home page in a grid and allow users to search products by product name.
Preconditions	User opens the EShop home page; product data is available from the product API or local seed data.
Main flow	System loads product data, displays loading state while waiting, renders product cards in a grid, user enters a search keyword, system filters products by name, and matching products remain visible.
Alternative flows	Product list is empty, search has no match, search keyword is empty/whitespace, product image is missing, product price is invalid, or keyword contains HTML/script-like input.
Related data	Product id, name, price, image, image alt text, and product list API/loading state.

2. Input / Output Identification

Inputs

No.	Input	Type	Required?	Source	Constraint / Rule	Notes
1	Product list data	Array/object	Yes	Product API or local seed data	Each product should include name, price, image, and alt text.	Drives the grid display.
2	Product name	Text	Yes	Product data	Must be displayed as text.	Used by search matching.

No.	Input	Type	Required?	Source	Constraint / Rule	Notes
3	Search keyword	Text	No	Search input	Search by product name; input must be rendered safely and must not execute HTML/JS.	Empty keyword should show all products.
4	Loading state	Boolean	Yes	Frontend/API state	Shows loading indicator while products are being fetched.	Must disappear after response.

Outputs

No.	Output	Trigger	Expected Behavior	Notes
1	Product grid	Product data loaded	Displays all products as cards in a responsive grid.	Cards should not overlap.
2	Safe keyword display	Keyword contains HTML/script-like text	Keyword is displayed as text only, never rendered as HTML.	Prevents XSS.
3	Loading state	Data request is pending	A loading message/spinner is shown.	Should not show stale empty state during loading.
4	Empty state	Search returns no matching products or product list is empty	Clear empty-state message is shown.	Should not show broken grid.

3. Domain Rule Identification

Rule ID	Field / Condition	Valid Rule	Invalid Condition	Error Message / Expected Handling
FR05-R01	Product grid	Loaded product data is rendered in a grid.	Products are missing, duplicated, or layout is broken.	Show correct product cards or an empty state if no products exist.
FR05-R02	Product image	Image keeps standard aspect ratio and has meaningful alt text.	Missing alt text, broken image, stretched image, or layout shift.	Show accessible image/fallback without breaking card layout.
FR05-R03	Product name	Product name is displayed as text.	Name is missing, clipped incorrectly, or rendered as HTML.	Escape text and keep layout readable.
FR05-R04	Product price	Price is displayed with thousands separators and đ.	Missing currency, wrong separator, raw number, negative/non-numeric value.	Display valid formatted price or handle invalid data safely.
FR05-R05	Search by name	Keyword filters by product name.	Search filters unrelated fields or returns incorrect products.	Show only matching product names.
FR05-R06	Empty keyword	Empty keyword shows all products.	Empty keyword hides all products or shows error.	Reset to full product list.
FR05-R07	No result	Keyword has no match.	Blank page or stale results remain visible.	Show appropriate empty-state message.
FR05-R08	Safe keyword rendering	User input is escaped.	HTML/script is executed or rendered.	Display input safely and do not execute scripts.
FR05-R09	Loading	Loading state appears while data is pending.	No loading state or incorrect empty state during fetch.	Show loading indicator until response completes.

4. Equivalence Class Partitioning

EC ID	Input / Condition	Class Type	Description	Representative Value	Expected Result
FR05-EC-008	Keyword matches exact product name	Valid	Search term equals a product name.	iPhone 15 Pro Max	Matching product is displayed.
FR05-EC-009	Keyword matches partial product name	Valid	Search term is a substring.	iPhone	Matching products are displayed.
FR05-EC-010	Keyword differs only by case	Valid	Keyword uses different letter case.	iphone	Matching products are displayed if search is case-insensitive.
FR05-EC-011	Keyword has no match	Valid	No product name contains keyword.	Nintendo	Empty state is displayed.
FR05-EC-012	Empty keyword	Valid	Search field is empty.	Empty string	All products are displayed.
FR05-EC-013	Whitespace-only keyword	Valid	Keyword contains only spaces.		Keyword is trimmed and all products are displayed.
FR05-EC-014	Keyword contains HTML/script	Invalid	User enters markup/script-like text.	<script>alert(1)</script>	Input is displayed safely and no script executes.
FR05-EC-015	Keyword contains special characters	Valid	Keyword includes punctuation.	Q1!@#	Search handles input safely and returns matching or empty result.
FR05-EC-016	Keyword exceeds expected max length	Invalid	Very long keyword.	256 characters	App handles safely without crash or layout break.

5. Domain Testing Test Cases

Test Case ID	Objective	Preconditions	Test Data	Steps	Expected Result	Actual Result	Status
FR05-DT-001	Verify product grid displays all products.	Product returns seeded product list.	Products: iPhone 15, Samsung S24, MacBook Pro, AirPods Pro, Keychron Q1.	Open home page.	All products are displayed in a grid with one card per product.	All products are displayed in a grid with one card per product.	Passed
FR05-DT-007	Verify exact-name search.	Product list is loaded.	Keyword MacBook Pro M3.	Enter keyword and submit search.	Only MacBook Pro M3 is displayed.	Only MacBook Pro M3 is displayed.	Passed
FR05-DT-008	Verify partial-name search.	Product list is loaded.	Keyword Samsung.	Enter keyword and submit search.	Matching Samsung product(s) are displayed.	Matching Samsung product(s) are displayed.	Passed
FR05-DT-009	Verify case-insensitive search.	Product list is loaded.	Keyword iphone.	Enter keyword and submit search.	iPhone 15 Pro Max is displayed if search is case-insensitive.	iPhone 15 Pro Max is displayed.	Passed
FR05-DT-010	Verify no-result search empty state.	Product list is loaded.	Keyword Nintendo.	Enter keyword and submit search.	No product cards are displayed and an appropriate empty-state message is shown.	No product cards are displayed.	Passed
FR05-DT-011	Verify empty search resets product list.	Product list is loaded.	Empty keyword.	Clear search input and submit search.	Full product list is displayed.	Full product list is displayed.	Passed
FR05-DT-012	Verify whitespace-only search is trimmed.	Product list is loaded.	Keyword .	Enter spaces and submit search.	App treats keyword as empty and displays full product list.	App treats keyword as empty and displays full product list.	Passed
FR05-DT-013	Verify search keyword is rendered safely.	Product list is loaded.	Keyword <script>alert(1)</script>.	Enter keyword and submit search.	Script is not executed; keyword is treated as plain text; safe empty state may be shown.	Script is not executed.	Passed

6. Boundary Identification

Boundary ID	Field / Condition	Boundary Rule	Values to Test	Reason
FR05-BD-002	Search keyword length minimum	Empty keyword is allowed and resets the list.	0, 1, 2 characters	Tests empty, minimum non-empty, and just above minimum.
FR05-BD-003	Search keyword length maximum	Assumed practical maximum is 255 characters.	254, 255, 256 characters	Verifies long input handling and UI stability.
FR05-BD-004	Search result count	Results can be none, one, or many.	0, 1, 2+ matches	Verifies empty state and filtered grid.

7. Boundary Value Analysis Test Cases

Test Case ID	Boundary Target	Preconditions	Test Data	Steps	Expected Result	Actual Result	Status
FR05-BVA-001	Product count 0	Product API can return empty list.	[]	Open home page.	Empty-state message is shown and no broken cards appear.	TBD	Not Executed
FR05-BVA-002	Product count 1	Product API can return one product.	One product only.	Open home page.	One product card is displayed cleanly in the grid area.	TBD	Not Executed
FR05-BVA-003	Product count 2+	Product API returns multiple products.	Five products.	Open home page.	Multiple product cards are displayed in grid layout.	TBD	Not Executed
FR05-BVA-004	Keyword length 0	Product list is loaded.	Empty keyword.	Submit empty search.	Full product list is displayed.	Full product list is displayed.	Passed
FR05-BVA-005	Keyword length 1	Product list is loaded.	Keyword Q.	Submit search.	Matching products are shown or empty state appears; app remains stable.	Matching products are shown.	Passed
FR05-BVA-006	Keyword length 255	Product list is loaded.	255-character keyword.	Paste keyword and submit search.	App handles input safely without crash or layout break.	App handles input safely without crash or layout break.	Passed
FR05-BVA-007	Keyword length 256	Product list is loaded.	256-character keyword.	Paste keyword and submit search.	App rejects, truncates, or safely handles keyword according to confirmed product rule.	App handles input safely without crash or layout break.	Passed

4. FR-09 Discount Coupons

Source: HW02/test-cases/FR09-discount-coupons.md

1. Feature Understanding

Item	Details
Feature ID	FR-09

Item	Details
Feature name	Discount Coupons
User role	Logged-in customer
Main purpose	Allow a customer to apply a coupon at checkout and reduce the payable order total when all coupon conditions are satisfied.
Preconditions	Customer is at Checkout; cart has at least one product; coupon data exists from FR-17 Coupon Management CRUD; customer has valid JWT token for authenticated scenarios.
Main flow	Customer enters coupon code, clicks Apply, system validates all coupon conditions, calculates discount, displays success message, discount amount, and final payable amount.
Alternative flows	Coupon does not exist, inactive, expired, below minimum order, user is not logged in, usage limit reached, or discount calculation creates invalid final amount.
Related data	Coupon records from FR-17: code, type, discount_value, expired_at, min_order_amount, max_uses_per_user, is_active; user coupon usage history; cart total.

2. Input / Output Identification

Inputs

No.	Input	Type	Required?	Source	Constraint / Rule	Notes
1	Coupon code	Text	Yes	Customer checkout form	Must exist in database and be active (is_active = 1)	Depends on FR-17 coupon records.
2	Cart total	Money/number	Yes	Cart/checkout state	Must be greater than or equal to coupon min_order_amount	Used for minimum threshold and discount formula.
3	User authentication token	JWT/session	Yes	Logged-in customer session	Must be valid	Required by C4.
4	Current date	Date	Yes	System date	Must be before expired_at	Project date: 2026-06-23.
5	Coupon active state	Boolean	Yes	Coupon database	is_active = 1	Inactive coupon requires setup from FR-17/test data.
6	Coupon usage count for user	Integer	Yes	Coupon usage history/order history	Must be less than max_uses_per_user	Boundary is max_uses_per_user - 1, max_uses_per_user, max_uses_per_user + 1.
7	Coupon type	Enum	Yes	Coupon database	percent or fixed	Created/managed through FR-17.
8	Coupon discount value	Number	Yes	Coupon database	Percent or fixed amount depending on type	Calculation source.

Outputs

No.	Output	Trigger	Expected Behavior	Notes
1	Success message	Valid coupon applied	Display coupon applied successfully.	Exact message text to verify from UI.
2	Error message	Invalid coupon condition	Display clear reason or generic rejection.	Must not apply discount.
3	Discount amount	Valid coupon applied	Calculate by coupon formula.	percent: total * value / 100; fixed: fixed value.
4	Final amount	Valid coupon applied	final_amount = total - discount_amount	Should not become negative; requirement needs clarification.
5	Payment total display	Coupon applied or rejected	Shows final amount after valid coupon or original total after invalid coupon.	Must be consistent across UI fields.
6	Usage record	Order/payment confirmed after coupon use	User coupon usage count increases after successful use.	Exact timing to verify: apply vs payment confirmation.

3. Domain Rule Identification

Rule ID	Field / Condition	Valid Rule	Invalid Condition	Error Message / Expected Handling
FR09-R01	C1 - Coupon exists and active	Code exists and is_active = 1.	Code does not exist or is_active = 0.	Reject coupon; do not change final amount.
FR09-R02	C2 - Not expired	Current date is before expired_at.	Current date is on/after expired date, depending on confirmed date semantics.	Reject coupon as expired.
FR09-R03	C3 - Minimum order	Cart total is greater than or equal to min_order_amount.	Cart total is below min_order_amount.	Reject coupon because order does not meet minimum.
FR09-R04	C4 - Logged in	User has valid JWT token/session.	User is logged out or token is missing/invalid.	Reject coupon or redirect/login required.
FR09-R05	C5 - Usage limit	User usage count is less than max_uses_per_user.	Usage count is equal to or greater than max_uses_per_user.	Reject coupon because usage limit reached.

Rule ID	Field / Condition	Valid Rule	Invalid Condition	Error Message / Expected Handling
FR09-R06	Percent calculation	<code>discount_amount = total * discount_value / 100.</code>	Discount amount is calculated incorrectly.	Show correct discount and final amount.
FR09-R07	Fixed calculation	<code>discount_amount = discount_value.</code>	Discount amount is calculated incorrectly.	Show correct discount and final amount.
FR09-R08	Final amount	<code>final_amount = total - discount_amount.</code>	Final amount is inconsistent or negative without defined behavior.	Clarify whether discount should be capped at cart total.
FR09-R09	FR-17 dependency	Coupon data from FR-17 is accurate and valid.	FR-17 allows invalid coupon data that FR-09 later applies.	Reject invalid coupon at FR-09 or record cross-feature defect.

4. Equivalence Class Partitioning

EC ID	Input / Condition	Class Type	Description	Representative Value	Expected Result
FR09-EC-001	Coupon exists and active	Valid	Existing active coupon.	SAVE10	Coupon can be applied if all other conditions pass.
FR09-EC-002	Coupon does not exist	Invalid	Unknown coupon code.	NO_SUCH_CODE	Coupon is rejected.
FR09-EC-003	Coupon inactive	Invalid	Existing coupon with <code>is_active = 0.</code>	INACTIVE10	Coupon is rejected.
FR09-EC-004	Coupon not expired	Valid	Expiration date in future.	SAVE10 expires 2099-12-31	Coupon can be applied if other conditions pass.
FR09-EC-005	Coupon expired	Invalid	Expiration date in past.	EXPIRED expires 2020-01-01	Coupon is rejected.
FR09-EC-006	Cart total meets minimum	Valid	Total is equal to or above threshold.	SAVE10 with 300000	Coupon can be applied if other conditions pass.
FR09-EC-007	Cart total below minimum	Invalid	Total is below threshold.	SAVE10 with 299999	Coupon is rejected.
FR09-EC-008	Logged-in user	Valid	User has valid JWT token.	Test User logged in	Coupon can be applied if other conditions pass.
FR09-EC-009	Logged-out user	Invalid	Missing/invalid token.	Guest user	Coupon is rejected or login required.
FR09-EC-010	Usage count below limit	Valid	User used coupon fewer times than limit.	VIP100 , used 1 of 2	Coupon can be applied.
FR09-EC-011	Usage limit reached	Invalid	User usage count equals limit.	SAVE10 , used 1 of 1	Coupon is rejected.
FR09-EC-012	Percent coupon	Valid	Coupon type is <code>percent.</code>	SAVE10	Discount is calculated by percentage formula.
FR09-EC-013	Fixed coupon	Valid	Coupon type is <code>fixed.</code>	BIGBUY	Discount is fixed amount.
FR09-EC-014	Fixed discount greater than cart total	Invalid / Requirement unclear	Fixed discount exceeds order total.	Test coupon FIXED_OVER_TOTAL , total 50000 , discount 100000	Requirement unclear; system should not produce negative final amount.

5. Domain Testing Test Cases

Test Case ID	Objective	Preconditions	Test Data	Steps	Expected Result	Actual Result	Status
FR09-DT-001	Verify valid percent coupon can be applied.	User is logged in; cart total meets minimum; SAVE10 active, not expired, usage not reached.	Cart total 28000000; coupon SAVE10	Open checkout, enter SAVE10 , click Apply.	Coupon is applied successfully; discount amount is 2800000; final amount is 25200000.	Counpon is applied but final amount remains unchanged	Failed

Test Case ID	Objective	Preconditions	Test Data	Steps	Expected Result	Actual Result	Status
FR09-DT-002	Verify invalid/unknown coupon is rejected.	User is logged in; cart has eligible product.	Coupon NO_SUCH_CODE	Open checkout, enter unknown coupon, click Apply.	Coupon is rejected; final amount remains unchanged; clear error message is shown.	Coupon is rejected; final amount remains unchanged; clear error message is shown.	Passed
FR09-DT-003	Verify expired coupon is rejected.	User is logged in; cart total meets EXPIRED minimum.	Cart total 100000; coupon EXPIRED; expired_at 2020-01-01.	Open checkout, enter EXPIRED, click Apply.	Coupon is rejected as expired; final amount remains unchanged.	Coupon is rejected as expired; final amount remains unchanged.	Passed
FR09-DT-005	Verify coupon usage limit reached is rejected.	User is logged in; user has already used SAVE10 once; SAVE10.max_uses_per_user=1.	Coupon SAVE10; usage count 1; max uses 1; cart total 300000.	Open checkout, enter SAVE10, click Apply.	Coupon is rejected because usage limit is reached; final amount remains unchanged.	Coupon is rejected because usage limit is reached; final amount remains unchanged.	Passed
FR09-DT-006	Verify cart total below minimum order rejects coupon.	User is logged in; FR09-DT-006 active and not expired.	Coupon FR09-DT-006; cart total 28000000; min order 30000000.	Open checkout with cart below threshold, enter FR09-DT-006, click Apply.	Coupon is rejected because cart total is below min_order_amount.	Coupon is rejected because cart total is below min_order_amount.	Passed
FR09-DT-007	Verify percentage discount calculation.	User is logged in; FR09-DT-007 can still be used by the user.	Cart total 28000000; coupon FR09-DT-007; value 10%.	Apply FR09-DT-007 at checkout.	Discount amount is 10%; final amount is 2520000.	Coupon is applied but final amount remains unchanged	Failed
FR09-DT-008	Verify fixed amount discount calculation.	User is logged in; BIGBUY active, not expired, usage not reached; cart meets minimum.	Cart total 28000000; coupon BIGBUY; fixed discount 50000.	Apply BIGBUY at checkout.	Discount amount is 50000; final amount is 27950000.	final amount is 27950000	Passed
FR09-DT-009	Verify fixed discount greater than cart total is handled safely.	User is logged in; test coupon exists with fixed discount greater than cart total.	Coupon FIXED_OVER_TOTAL; cart total 50000; fixed discount 100000.	Apply FIXED_OVER_TOTAL at checkout.	Requirement unclear: system should not produce negative final amount; expected behavior should be reject coupon or cap final amount at 0.	System produce negative final amount	Failed
FFR09-DT-010	Verify logged-out user cannot apply coupon.	User is logged out or JWT token is missing/invalid.	Coupon SAVE10; cart total 300000.	Open checkout as guest or remove token, enter SAVE10, click Apply.	Coupon is not applied; login required or unauthorized message is shown.	TBD	Not Executed

6. Boundary Identification

Boundary ID	Field / Condition	Boundary Rule	Values to Test	Reason
FR09-BD-001	Cart total vs min_order_amount for SAVE10	Coupon valid when total >= 300000.	299999, 300000, 300001	Tests just below, exactly at, and just above minimum order.
FR09-BD-002	Cart total vs min_order_amount for BIGBUY	Coupon valid when total >= 500000.	499999, 500000, 500001	Tests fixed coupon threshold.
FR09-BD-003	Usage limit for SAVE10	Valid when used count < 1.	0, 1, 2	Tests below limit, exactly at limit, and above limit.
FR09-BD-004	Usage limit for VIP100	Valid when used count < 2.	1, 2, 3	Tests below limit, at limit, and above limit for max uses 2.
FR09-BD-005	Expiration date boundary	Current date must be before expired_at.	2026-06-22, 2026-06-23, 2026-06-24	Tests before current date, exactly current date, and after current date with project date 2026-06-23.
FR09-BD-006	Percent discount calculation	Percent discount should match formula exactly.	total=1000000, discount_value=10	Verifies expected discount 100000.
FR09-BD-007	Fixed discount equal/greater than total	Final amount should not be unsafe.	fixed 49999, 50000, 50001 against total 50000	Tests just below, equal to, and greater than cart total.
FR09-BD-008	Coupon code existence boundary	Code field must map to an existing active coupon.	Empty, unknown code, existing code	Separates missing/invalid/existing coupon classes.

Test Case ID	Boundary Target	Preconditions	Test Data	Steps	Expected Result	Actual Result	Status
FR09-BVA-001	Cart total just below VIP100 minimum	User logged in; VIP100 active, not expired, usage not reached.	Cart total 299999; coupon SAVE10; min 300000.	Apply SAVE10 at checkout.	Coupon is rejected because total is below minimum.	Coupon is rejected because total is below minimum.	Passed
FR09-BVA-002	Cart total exactly at VIP100 minimum	User logged in; VIP100 active, not expired, usage not reached.	Cart total 300000; coupon VIP100.	Apply VIP100 at checkout.	Coupon is accepted; discount is 30000; final amount is 270000.	Coupon is rejected because total is below minimum.	Failed
FR09-BVA-003	Cart total just above VIP100 minimum	User logged in; VIP100 active, not expired, usage not reached.	Cart total 300001; coupon VIP100.	Apply VIP100 at checkout.	Coupon is accepted; discount is 30000.1 before rounding; rounding/display rule must be verified.	Coupon is accepted	Passed
FR09-BVA-007	Usage count below SAVE10 limit	User logged in; usage history can be prepared.	SAVE10; used count 0; max uses 1.	Apply SAVE10.	Coupon is accepted if all other conditions pass.	Coupon is accepted	Passed
FR09-BVA-008	Usage count exactly at SAVE10 limit	User logged in; usage history can be prepared.	SAVE10; used count 1; max uses 1.	Apply SAVE10.	Coupon is rejected because usage count is not less than max uses.	Coupon is rejected	Passed
FR09-BVA-009	Usage count below VIP100 limit	User logged in; usage history can be prepared.	VIP100; used count 1; max uses 2.	Apply VIP100.	Coupon is accepted	Coupon is accepted	Passed
FR09-BVA-010	Expiration exactly at current date	User logged in; coupon exists with expired_at=2026-06-30.	Coupon EXPIRES_TODAY; current date 2026-06-30.	Apply coupon.	Requirement unclear: coupon should be rejected unless end-of-day behavior is defined.	Coupon is rejected	Passed
FR09-BVA-011	Expiration after current date	User logged in; coupon exists with expired_at=2026-07-01.	Coupon EXPIRES_TOMORROW; current date 2026-07-01.	Apply coupon.	Coupon is accepted if all other conditions pass.	Coupon is accepted	Passed
FR09-BVA-012	Fixed discount equals cart total	User logged in; test fixed coupon exists.	Cart total 50000; fixed discount 50000.	Apply coupon.	Final amount is 0; payment flow handles zero amount according to product rule.	Final amount is 0; payment flow handles zero amount according to product rule.	Passed
FR09-BVA-013	Fixed discount greater than cart total	User logged in; test fixed coupon exists.	Cart total 50000; fixed discount 50001.	Apply coupon.	Final amount is 0; payment flow handles zero amount according to product rule.	Final amount is negative value	Failed

5. FR-17 Coupon Management CRUD

Source: HW02/test-cases/FR17-coupon-management-crud.md

1. Feature Understanding

Item	Details
Feature ID	FR-17
Feature name	Coupon Management CRUD
User role	Admin
Main purpose	Allow an admin to add, view, and delete discount coupons.
Preconditions	Admin is authenticated and has access to the EShop Admin coupon management page.
Main flow	Admin opens Coupon Management, enters coupon data, creates a coupon, sees the coupon in the list, and can delete an existing coupon.
Alternative flows	Validation error for missing/invalid input; duplicate coupon code is rejected; expired coupon is displayed as expired; delete may require confirmation.
Related data	Coupon code, type, discount value, expiration date, minimum order amount, maximum uses per user. FR-17 data is consumed by FR-09 Discount Coupons.

2. Input / Output Identification

Inputs

No.	Input	Type	Required?	Source	Constraint / Rule	Notes
1	code	Text	Yes	Admin create coupon form	Required and unique	UI should trim or reject whitespace-only input; max length must be verified.
2	type	Enum	Yes	Admin create coupon form	Allowed values shown by UI: percent , fixed	UI labels may display as percentage/fixed in Vietnamese.
3	discount_value	Number	Yes	Admin create coupon form	Must be positive	Percent upper bound is assumed to be 100. UI numeric validation is in scope.
4	expired_at	Date	Yes	Admin create coupon form	Required	Valid date range must be verified through the form.
5	min_order_amount	Money/number	Yes	Admin create coupon form	Must be >= 0	Currency appears to be VND. UI numeric validation is in scope.
6	max_uses_per_user	Integer	Yes	Admin create coupon form	Must be >= 1	UI should reject decimal values or prevent them.
7	Delete action	Button/action	Yes for delete flow	Coupon table row	Existing coupon selected for deletion	Confirmation behavior must be verified.
8	Coupon list state	Existing data	Yes for view/delete flows	System database/seed data	Coupon records can be active or expired	Existing codes shown in screenshot include SAVE10 , BIGBUY , VIP100 , and EXPIRED .

Outputs

No.	Output	Trigger	Expected Behavior	Notes
1	Created coupon row	Admin submits valid create form	New coupon appears in the coupon list with correct values.	Exact success message is not specified.
2	Validation error	Admin submits missing or invalid fields	Coupon is not created and field/form error is shown.	Exact message text must be verified.
3	Duplicate code error	Admin submits an existing coupon code	Coupon is not created and duplicate code error is shown.	Use existing code such as SAVE10 only if present in the test environment.
4	Coupon list	Admin opens coupon management page	Existing coupons are displayed with code, type, value, minimum order, expiration, uses/user, and action.	View/read behavior.
5	Expired status	Coupon expiration date is in the past	Coupon is displayed as expired or otherwise unavailable.	Screenshot shows an expired row label.
6	Deleted coupon state	Admin deletes an existing coupon	Coupon is removed from list or marked deleted/unavailable.	Exact behavior must be verified.

3. Domain Rule Identification

Rule ID	Field / Condition	Valid Rule	Invalid Condition	Error Message / Expected Handling
FR17-R01	Admin access	Authenticated admin can access coupon management.	Unauthenticated or non-admin user accesses page.	Access is denied or redirected.
FR17-R02	code required	Non-empty coupon code is provided.	Empty or whitespace-only code.	Reject creation and show required-field validation.
FR17-R03	code unique	Code does not already exist.	Code duplicates an existing coupon.	Reject creation and show duplicate-code validation.
FR17-R04	type required/allowed	Type is one of the values available in the admin form, normally percent or fixed.	Missing type selection.	Reject creation and show type validation.
FR17-R05	discount_value required	Discount value is provided.	Empty value.	Reject creation and show required-field validation.

Rule ID	Field / Condition	Valid Rule	Invalid Condition	Error Message / Expected Handling
FR17-R06	discount_value positive	Discount value is greater than 0.	Zero or negative value.	Reject creation and show positive-value validation.
FR17-R07	Percent discount range	Percent discount is assumed valid from 1 to 100.	Percent value greater than 100.	Reject creation or cap behavior must be verified.
FR17-R08	expired_at required	Expiration date is provided.	Empty expiration date.	Reject creation and show required-field validation.
FR17-R09	Expiration date active rule	Assumed valid when date is today or future for a new active coupon.	Past expiration date for a new coupon.	Reject creation or create as expired; expected behavior must be verified.
FR17-R10	min_order_amount required	Minimum order amount is provided.	Empty value.	Reject creation and show required-field validation.
FR17-R11	min_order_amount lower bound	Minimum order amount is greater than or equal to 0.	Negative value.	Reject creation and show minimum-value validation.
FR17-R12	max_uses_per_user required	Maximum uses per user is provided.	Empty value.	Reject creation and show required-field validation.
FR17-R13	max_uses_per_user lower bound	Maximum uses per user is an integer greater than or equal to 1.	Zero, negative, or decimal value.	Reject creation and show usage-limit validation.
FR17-R14	View coupon list	Existing coupons are readable by admin.	List fails to load or omits saved coupon.	Show correct list or error state.
FR17-R15	Delete coupon	Existing coupon can be deleted by admin from the coupon table.	Delete action fails for a visible existing coupon.	Coupon remains visible or system shows a clear delete error.

4. Equivalence Class Partitioning

EC ID	Input / Condition	Class Type	Description	Representative Value	Expected Result
FR17-EC-001	Admin role	Valid	Authenticated admin opens coupon management.	Admin user	Page displays coupon form and list.
FR17-EC-002	Admin role	Invalid	Non-admin or unauthenticated user opens coupon management.	Guest user	Access is denied or redirected.
FR17-EC-003	code	Valid	New non-empty unique code.	NEW10	Coupon can be created when other fields are valid.
FR17-EC-004	code	Invalid	Empty code.	Empty string	Creation is rejected.
FR17-EC-005	code	Invalid	Duplicate code.	SAVE10	Creation is rejected as duplicate.
FR17-EC-006	type	Valid	Percent type.	percent	Coupon can be created when other fields are valid.
FR17-EC-007	type	Valid	Fixed type.	fixed	Coupon can be created when other fields are valid.
FR17-EC-008	type	Invalid	No type selected in the admin form.	Empty type	Creation is rejected.
FR17-EC-009	discount_value	Valid	Positive percent discount within assumed range.	10	Coupon can be created when other fields are valid.
FR17-EC-010	discount_value	Valid	Positive fixed discount.	50000	Coupon can be created when other fields are valid.
FR17-EC-011	discount_value	Invalid	Missing discount value.	Empty	Creation is rejected.
FR17-EC-012	discount_value	Invalid	Zero discount.	0	Creation is rejected.
FR17-EC-013	discount_value	Invalid	Negative discount.	-1	Creation is rejected.
FR17-EC-014	discount_value percent	Invalid	Percent discount above assumed maximum.	101	Creation is rejected or behavior must be verified.
FR17-EC-015	expired_at	Valid	Required date in the future.	2099-12-31	Coupon can be created when other fields are valid.
FR17-EC-016	expired_at	Invalid	Missing expiration date.	Empty	Creation is rejected.
FR17-EC-017	expired_at	Invalid	Past expiration date for newly created active coupon.	2020-01-01	Creation is rejected or created as expired; behavior must be verified.
FR17-EC-018	min_order_amount	Valid	Zero minimum order amount.	0	Coupon can be created when other fields are valid.
FR17-EC-019	min_order_amount	Valid	Positive minimum order amount.	300000	Coupon can be created when other fields are valid.
FR17-EC-020	min_order_amount	Invalid	Missing minimum order amount.	Empty	Creation is rejected.
FR17-EC-021	min_order_amount	Invalid	Negative minimum order amount.	-1	Creation is rejected.

EC ID	Input / Condition	Class Type	Description	Representative Value	Expected Result
FR17-EC-022	max_uses_per_user	Valid	Integer at least 1.	1	Coupon can be created when other fields are valid.
FR17-EC-023	max_uses_per_user	Invalid	Zero uses per user.	0	Creation is rejected.
FR17-EC-024	max_uses_per_user	Invalid	Negative uses per user.	−1	Creation is rejected.
FR17-EC-025	max_uses_per_user	Invalid	Decimal uses per user.	1.5	Creation is rejected.
FR17-EC-026	Coupon list	Valid	Existing coupons are displayed.	Seed coupon list	List shows coupon details.
FR17-EC-027	Delete existing coupon	Valid	Admin deletes an existing coupon.	NEW10	Coupon is removed or unavailable after delete.
FR17-EC-028	Delete existing coupon	Invalid	Delete action for a visible coupon fails or does not update the list.	Visible coupon row	System keeps list consistent and shows a clear error if delete fails.

Additional Equivalence Classes

EC ID	Input / Condition	Class Type	Description	Representative Value	Expected Result
FR17-EC-029	code whitespace-only	Invalid	Covers FR17-DT-018: code contains only spaces and should be treated as empty after trimming.	" "	Coupon is rejected with required/invalid code validation and coupon is not created.
FR17-EC-030	code leading/trailing spaces	Invalid	Covers FR17-DT-019: code contains spaces around an existing coupon code.	" FR17-DT-019 "	Inconclusive - Requirement unclear: system should trim before uniqueness check or reject untrimmed input; it must not create a confusing duplicate.
FR17-EC-031	code special separator character	Valid	Covers FR17-DT-020: coupon code contains a separator character and allowed characters are not specified.	SAVE_10	Inconclusive - Requirement unclear because allowed coupon code characters are not specified.
FR17-EC-032	code Unicode/Vietnamese characters	Valid	Covers FR17-DT-021: coupon code contains Vietnamese diacritics.	GIÃM10	Inconclusive - Requirement unclear because Unicode coupon code support is not specified.

5. Domain Testing Test Cases

Test Case ID	Objective	Preconditions	Test Data	Steps	Expected Result	Actual Result	Status
FR17-DT-001	Verify admin can view coupon management page and coupon list.	Admin account exists and is logged in.	Existing coupons available.	Open EShop Admin > Ma Giam Gia.	Coupon form and coupon table are displayed with coupon fields.	Coupon form and coupon table are displayed with coupon fields.	Passed
FR17-DT-002	Verify admin can create a valid percent coupon.	Admin is logged in; code NEW10 does not exist.	code=NEW10; type=percent; discount_value=10; expired_at=2099-12-31; min_order_amount=0; max_uses_per_user=1	Fill all fields and submit create coupon form.	Coupon is created and appears in the list with matching values.	Coupon is created and appears in the list with matching values.	Passed
FR17-DT-003	Verify admin can create a valid fixed coupon.	Admin is logged in; code FIXED50K does not exist.	code=FIXED50K; type=fixed; discount_value=50000; expired_at=2099-12-31; min_order_amount=300000; max_uses_per_user=2	Fill all fields and submit create coupon form.	Coupon is created and appears in the list with matching values.	Coupon is created and appears in the list with matching values.	Passed
FR17-DT-004	Verify empty coupon code is rejected.	Admin is logged in.	code= empty; other fields valid.	Leave code blank and submit create coupon form.	Coupon is not created and code required validation is shown.	Coupon is not created and code required validation is shown.	Passed
FR17-DT-005	Verify duplicate coupon code is rejected.	Admin is logged in; coupon SAVE10 exists.	code=SAVE10; other fields valid.	Enter existing code and submit create coupon form.	Coupon is not created and duplicate code validation is shown.	Coupon is not created and duplicate code validation is shown.	Passed
FR17-	Verify missing discount value is rejected.	Admin is logged in.	discount_value= empty; other fields valid.	Leave discount value blank and	Coupon is not created and discount required validation is shown.	Coupon is not created and discount required validation is shown.	Passed

Test Case ID	Objective	Preconditions	Test Data	Steps	Expected Result	Actual Result	Status
DT-006				submit create coupon form.			
FR17-DT-007	Verify zero discount value is rejected.	Admin is logged in.	discount_value=0; other fields valid.	Enter zero discount value and submit create coupon form.	Coupon is not created and positive discount validation is shown.	Coupon is created	Failed
FR17-DT-008	Verify negative discount value is rejected.	Admin is logged in.	discount_value=-1; other fields valid.	Enter negative discount value and submit create coupon form.	Coupon is not created and positive discount validation is shown.	Coupon is created	Failed
FR17-DT-009	Verify percent discount above assumed maximum is rejected.	Admin is logged in; percent upper limit confirmed as 100 or test marked for review.	type=percent; discount_value=101; other fields valid.	Enter percent discount over 100 and submit create coupon form.	Coupon is not created or the observed behavior is flagged for rule clarification.	Coupon is created	Failed
FR17-DT-010	Verify missing expiration date is rejected.	Admin is logged in.	expired_at= empty; other fields valid.	Leave expiration date blank and submit create coupon form.	Coupon is not created and expiration required validation is shown.	Coupon is not created and expiration required validation is shown.	Passed
FR17-DT-011	Verify past expiration date handling for new coupon.	Admin is logged in; expected active-coupon date rule needs confirmation.	expired_at=2020-01-01; other fields valid.	Enter past expiration date and submit create coupon form.	Coupon is rejected or created as expired according to confirmed requirement.	Coupon is created	Failed
FR17-DT-012	Verify missing minimum order amount is rejected.	Admin is logged in.	min_order_amount= empty; other fields valid.	Leave minimum order blank and submit create coupon form.	Coupon is not created and minimum order required validation is shown.	Coupon is created	Failed
FR17-DT-013	Verify negative minimum order amount is rejected.	Admin is logged in.	min_order_amount=-1; other fields valid.	Enter negative minimum order and submit create coupon form.	Coupon is not created and minimum order validation is shown.	Coupon is created	Failed
FR17-DT-014	Verify missing max uses per user is rejected.	Admin is logged in.	max_uses_per_user= empty; other fields valid.	Leave max uses per user blank and submit create coupon form.	Coupon is not created and max uses required validation is shown.	Coupon is not created and max uses required validation is shown.	Passed
FR17-DT-015	Verify zero max uses per user is rejected.	Admin is logged in.	max_uses_per_user=0; other fields valid.	Enter zero max uses per user and submit create coupon form.	Coupon is not created and max uses lower-bound validation is shown.	Coupon is not created	Passed
FR17-DT-016	Verify decimal max uses per user is rejected.	Admin is logged in.	max_uses_per_user=1.5; other fields valid.	Enter decimal max uses per user and submit create coupon form.	Coupon is not created and integer validation is shown.	Coupon is not created and integer validation is shown.	Passed
FR17-DT-017	Verify admin can delete an existing coupon.	Admin is logged in; test coupon NEW10 exists and is safe to delete.	code=NEW10	Click delete for the coupon and confirm if prompted.	Coupon is removed from the list or marked unavailable according to confirmed delete behavior.	Coupon is removed from the list or marked unavailable according to confirmed delete behavior.	Passed

Test Case ID	Objective	Preconditions	Test Data	Steps	Expected Result	Actual Result	Status
FR17-DT-018	Verify whitespace-only coupon code is rejected.	Admin is logged in.	code=" "; other fields valid.	Submit create coupon form.	Coupon is rejected because code is empty after trimming; coupon is not created.	Coupon is created	Failed
FR17-DT-019	Verify leading/trailing spaces are handled safely.	Admin is logged in; coupon FR17-DT-019 exists.	code=" FR17-DT-019 "; other fields valid.	Submit create coupon form.	Inconclusive - Requirement unclear: system should trim before uniqueness check or reject untrimmed code; it must not create a confusing duplicate.	Coupon is created	Inconclusive - Requirement unclear; potential defect
FR17-DT-020	Verify separator character in coupon code.	Admin is logged in; code SAVE_10 does not exist.	code=SAVE_10; other fields valid.	Submit create coupon form.	Inconclusive - Requirement unclear because allowed characters are not specified.	Coupon is created	Inconclusive - Requirement unclear; potential defect
FR17-DT-021	Verify Unicode/Vietnamese coupon code handling.	Admin is logged in; code GIÃM10 does not exist.	code=GIÃM10; other fields valid.	Submit create coupon form.	Inconclusive - Requirement unclear because Unicode coupon code support is not specified.	Coupon is created	Inconclusive - Requirement unclear; potential defect

6. Boundary Identification

Boundary ID	Field / Condition	Boundary Rule	Values to Test	Reason
FR17-BD-001	discount_value lower bound	Must be positive.	0, 1, 2	Tests min - 1, min, and min + 1 when minimum valid value is 1.
FR17-BD-002	Percent discount_value upper bound	Assumed maximum is 100.	99, 100, 101	Tests max - 1, max, and max + 1 for percent discounts.
FR17-BD-003	Fixed discount_value lower money boundary	Must be positive.	-1, 0, 1	Covers negative value, zero, and minimum valid value for money.
FR17-BD-004	min_order_amount lower bound	Must be greater than or equal to 0.	-1, 0, 1	Covers negative value, zero, and just above minimum.
FR17-BD-005	max_uses_per_user lower bound	Must be integer greater than or equal to 1.	0, 1, 2	Tests min - 1, min, and min + 1.
FR17-BD-006	max_uses_per_user integer rule	Must be whole number.	0.9, 1, 1.1	Tests just below/at/just above integer boundary.
FR17-BD-007	expired_at active date boundary	Assumed valid from the current date onward for new active coupons.	2026-06-21, 2026-06-22, 2026-06-23	Tests before boundary, exactly at boundary, and after boundary using the project date 2026-06-22.
FR17-BD-008	code length lower boundary	Required non-empty code.	Empty, 1 character, 2 characters	Tests missing, min, and min + 1 if minimum valid length is 1.
FR17-BD-009	code length upper boundary	Maximum length is not specified.	max - 1, max, max + 1 after requirement is confirmed	Boundary cannot be finalized until max length is known.

7. Boundary Value Analysis Test Cases

Test Case ID	Boundary Target	Preconditions	Test Data	Steps	Expected Result	Actual Result	Status
FR17-BVA-001	discount_value at lower invalid boundary	Admin is logged in.	type=percent; discount_value=0; other fields valid.	Submit create coupon form.	Coupon is rejected because discount value must be positive.	Coupon is created.	Failed

Test Case ID	Boundary Target	Preconditions	Test Data	Steps	Expected Result	Actual Result	Status
FR17-BVA-002	discount_value at minimum valid boundary	Admin is logged in.	type=percent; discount_value=1; other fields valid.	Submit create coupon form.	Coupon is created.	Coupon is created.	Passed
FR17-BVA-003	discount_value just above minimum	Admin is logged in.	type=percent; discount_value=2; other fields valid.	Submit create coupon form.	Coupon is created.	Coupon is created.	Passed
FR17-BVA-004	Percent discount just below assumed maximum	Admin is logged in.	type=percent; discount_value=99; other fields valid.	Submit create coupon form.	Coupon is created if percent max is 100.	Coupon is created.	Passed
FR17-BVA-005	Percent discount at assumed maximum	Admin is logged in.	type=percent; discount_value=100; other fields valid.	Submit create coupon form.	Coupon is created if percent max is 100.	Coupon is created	Passed
FR17-BVA-006	Percent discount above assumed maximum	Admin is logged in.	type=percent; discount_value=101; other fields valid.	Submit create coupon form.	Coupon is rejected or requirement gap is recorded if accepted.	Coupon is created	Failed
FR17-BVA-007	Fixed discount negative money boundary	Admin is logged in.	type=fixed; discount_value=-1; other fields valid.	Submit create coupon form.	Coupon is rejected because fixed discount must be positive.	Coupon is created	Failed
FR17-BVA-008	Fixed discount zero boundary	Admin is logged in.	type=fixed; discount_value=0; other fields valid.	Submit create coupon form.	Coupon is rejected because fixed discount must be positive.	Coupon is created	Failed
FR17-BVA-009	Fixed discount minimum valid money boundary	Admin is logged in	type=fixed; discount_value=1; other fields valid.	Submit create coupon form.	Coupon is created.	Coupon is created	Pass
FR17-BVA-010	min_order_amount below lower boundary	Admin is logged in.	min_order_amount=-1; other fields valid.	Submit create coupon form.	Coupon is rejected because minimum order cannot be negative.	Coupon is created	Failed
FR17-BVA-011	min_order_amount at lower boundary	Admin is logged in	min_order_amount=0; other fields valid.	Submit create coupon form.	Coupon is created.	Coupon is created	Passed
FR17-BVA-012	min_order_amount just above lower boundary	Admin is logged in.	min_order_amount=1; other fields valid.	Submit create coupon form.	Coupon is created.	Coupon is created	Passed
FR17-BVA-013	max_uses_per_user below lower boundary	Admin is logged in.	max_uses_per_user=0; other fields valid.	Submit create coupon form.	Coupon is rejected because max uses per user must be at least 1.	Coupon is created	Failed
FR17-BVA-014	max_uses_per_user at lower boundary	Admin is logged in.	max_uses_per_user=1; other fields valid.	Submit create coupon form.	Coupon is created.	Coupon is created	Passed
FR17-BVA-015	max_uses_per_user just above lower boundary	Admin is logged in.	max_uses_per_user=2; other fields valid.	Submit create coupon form.	Coupon is created.	Coupon is created	Passed
FR17-BVA-016	max_uses_per_user decimal below/near boundary	Admin is logged in.	max_uses_per_user=1.1; other fields valid.	Submit create coupon form.	Coupon is rejected because max uses per user must be an integer.	Coupon is created	Failed

Test Case ID	Boundary Target	Preconditions	Test Data	Steps	Expected Result	Actual Result	Status
FR17-BVA-017	expired_at before active-date boundary	Admin is logged in.	expired_at=2026-06-21; other fields valid.	Submit create coupon form.	Coupon is rejected or created as expired according to confirmed requirement.	Coupon is created	Failed
FR17-BVA-018	expired_at exactly at active-date boundary	Admin is logged in; code TODAY01 does not exist.	expired_at=2026-06-22; other fields valid.	Submit create coupon form.	Coupon is created if same-day expiration is allowed.	Coupon is created	Passed
FR17-BVA-019	expired_at after active-date boundary	Admin is logged in; code TOMOR01 does not exist.	expired_at=2026-06-23; other fields valid.	Submit create coupon form.	Coupon is created.	Coupon is created	Passed
FR17-BVA-020	code empty lower boundary	Admin is logged in.	code= empty; other fields valid.	Submit create coupon form.	Coupon is rejected because code is required.	Coupon is rejected because code is required.	Passed
FR17-BVA-021	code minimum non-empty boundary	Admin is logged in; code A does not exist; min code length confirmed as 1.	code=A; other fields valid.	Submit create coupon form.	Coupon is created if one-character codes are allowed; otherwise record requirement gap.	Coupon is created	Passed
FR17-BVA-022	code max length boundary placeholder	Admin is logged in; max code length is confirmed before execution.	code=max+1 length; other fields valid.	Submit create coupon form.	Coupon is rejected if it exceeds confirmed max length.	Coupon was created successfully with a 256-character code.	Inconclusive - max length is not specified in requirement.
FR17-BVA-023	code empty lower boundary	Admin is logged in.	code="" empty; other fields valid.	Submit create coupon form.	Coupon is rejected because code is required.	Coupon is rejected because code is required.	Passed

6. Mobile-FR04 Profile Management

Source: *HW02/test-cases/Mobile-FR04-profile-management.md*

1. Feature Understanding

Item	Details
Feature ID	Mobile-FR04
Feature name	Profile Management
User role	Logged-in customer
Main purpose	Allow a logged-in user to view/update their own profile information and review their order history from the mobile profile screen.
Preconditions	User is authenticated and opens the EShop Mobile profile page.
Main flow	User opens profile, reviews email, edits full name, phone number, and default shipping address, taps Cập nhật to save, and views Lịch sử đơn hàng.
Alternative flows	Invalid phone number is rejected; email remains non-editable; user cannot update another account; user cannot self-change role; order history can be empty; pending orders can show a cancel action; user logs out by tapping Thoát.
Related data	Email, full name, phone number, default shipping address, authenticated user ID/session, user role, order ID, order date, total amount, order status, and cancel action.

2. Input / Output Identification

Inputs

No.	Input	Type	Required?	Source	Constraint / Rule	Notes
1	Email	Text	Yes for display	Existing user profile	Non-editable through UI	Screenshot shows email field disabled/read-only.
2	Full name	Text	Yes	Profile update form	User can update own full name	Length and character rules must be verified.
3	Phone number	Text	No/conditional	Profile update form	If provided, starts with 0, digits only, length 10-11	Placeholder example: 0912345678.
4	Default shipping address	Textarea	No/conditional	Profile update form	User can update own default shipping address	Requiredness and max length must be verified.
5	Update action	Button/action	Yes for save flow	Cập nhật button	Saves valid editable fields for current user only	Must not change email or role.
6	Logout action	Button/action	Yes for logout flow	Thoát button	Ends current user session	Not the main update flow but visible on screen.

No.	Input	Type	Required?	Source	Constraint / Rule	Notes
7	Authenticated user identity	Session/context	Yes	Login/session state	User can update only their own profile	Prevents cross-account profile changes.
8	Role attribute	Hidden/system field	No UI input	System/user record	User cannot self-change role	Must not be accepted from profile UI data.
9	Order history data	Array/object	No	Current user's orders	Each row should belong to the logged-in user	Includes order number, date, total, and status.
10	Cancel order action	Button/action	Conditional	Pending order card	Available only for cancellable orders	Screenshot shows Hủy đơn for Chờ xác nhận.

Outputs

No.	Output	Trigger	Expected Behavior	Notes
1	Updated profile data	User submits valid editable fields	Full name, phone number, and default shipping address are saved for current user.	Email and role stay unchanged.
2	Validation error	User submits invalid phone/name/address data	Profile is not updated and a clear validation message is shown.	Exact message text must be verified.
3	Read-only email field	User views profile screen	Email is displayed but cannot be edited through UI.	Field may be disabled or read-only.
4	Authorization protection	User attempts to update another user's profile	Update is rejected and target account is unchanged.	May require API/devtools setup to verify beyond UI.
5	Role protection	User attempts to self-change role	Role remains unchanged and request is rejected/ignored.	Role control must not be exposed on profile screen.
6	Logout state	User taps Thoát	User session ends and user is redirected/logged out.	Secondary visible behavior.
7	Order history list	User opens profile page	User's orders are displayed under Lịch sử đơn hàng.	Screenshot shows Đơn #1.
8	Empty order state	User has no orders	Clear no-order message is displayed.	Earlier screenshot showed no orders.
9	Cancel order result	User cancels an eligible pending order	Order is canceled or status/list updates according to confirmed rule.	Exact confirmation behavior must be verified.

3. Domain Rule Identification

Rule ID	Field / Condition	Valid Rule	Invalid Condition	Error Message / Expected Handling
MFR04-R01	Authenticated access	Logged-in user can open profile page.	Guest opens profile page.	Redirect to login or deny access.
MFR04-R02	Own profile update	User updates only their own profile.	User attempts to update another user's profile.	Reject update and do not modify target user.
MFR04-R03	Email immutability	Email is displayed as non-editable.	User changes email through UI.	UI prevents editing; saved profile keeps original email.
MFR04-R04	Role immutability	Role is not exposed and cannot be changed by user.	User self-changes role.	Reject/ignore role change and keep original role.
MFR04-R05	Full name editable	Non-empty full name is accepted.	Full name is empty or whitespace-only.	Reject update and show required/invalid name validation.
MFR04-R06	Phone number optional/valid	Empty phone is allowed if optional; provided phone starts with 0 , digits only, length 10-11.	Phone is too short, too long, lacks leading 0 , contains letters/symbols/spaces.	Reject update and show phone validation.
MFR04-R07	Address editable	Default shipping address can be updated according to requiredness rule.	Address violates requiredness or length rule.	Reject update or save according to confirmed rule.
MFR04-R08	Successful update	Valid editable fields are saved.	Save fails silently or old values remain after refresh.	Show success state and persist updated values.
MFR04-R09	Logout	User can log out from profile page.	Logout button does not end session.	End session and redirect/update UI.
MFR04-R10	Order history visibility	Logged-in user sees their own order history.	Orders from another user are shown or own orders are missing.	Show only current user's orders or a clear empty state.
MFR04-R11	Order fields	Each order displays order number, order date, total amount, and status.	Required order fields are missing or unreadable.	Display complete order summary.
MFR04-R12	Order amount format	Total amount is formatted in VND with thousands separators and đ .	Raw number, wrong separator, missing currency, negative total.	Display valid formatted amount or handle invalid data.
MFR04-R13	Order status display	Order status is displayed in Vietnamese and matches stored order state.	Wrong status text or stale status is displayed.	Display correct status such as Chờ xác nhận .

Rule ID	Field / Condition	Valid Rule	Invalid Condition	Error Message / Expected Handling
MFR04-R14	Cancel pending order	Cancellable pending orders expose Hủy đơn.	Cancel button appears for non-cancellable orders or is missing for pending order.	Show/hide cancel action according to order status.

4. Equivalence Class Partitioning

EC ID	Input / Condition	Class Type	Description	Representative Value	Expected Result
MFR04-EC-001	Authenticated user	Valid	Logged-in user opens profile.	test@eshop.com session	Profile page is displayed.
MFR04-EC-002	Guest user	Invalid	Unauthenticated user opens profile.	No session	Access is denied or redirected to login.
MFR04-EC-003	Full name	Valid	Non-empty normal name.	Test User	Profile can be updated when other fields are valid.
MFR04-EC-004	Full name	Valid	Vietnamese name with diacritics.	Nguyen Van A / Nguyen Thi An	Profile can be updated if Unicode names are supported.
MFR04-EC-005	Full name	Invalid	Empty full name.	Empty string	Update is rejected.
MFR04-EC-006	Full name	Invalid	Whitespace-only full name.	" "	Update is rejected after trimming.
MFR04-EC-007	Phone number	Valid	9 digits according to executed system behavior.	123456789	Profile can be updated, but this conflicts with the stated 10-11 digit requirement.
MFR04-EC-008	Phone number	Valid	10 digits according to executed system behavior.	1912345678	Profile can be updated, but missing leading 0 conflicts with the stated requirement.
MFR04-EC-009	Phone number	Valid	Empty phone when optional.	Empty string	Profile can be updated if phone is optional.
MFR04-EC-010	Phone number	Invalid	8 digits, below observed lower boundary.	12345678	Update is rejected.
MFR04-EC-011	Phone number	Invalid	11 digits, above observed upper boundary.	12345678901	Update is rejected by the current system, but this conflicts with the stated 10-11 digit requirement.
MFR04-EC-012	Phone number	Invalid	10 digits starting with 0 rejected by current system.	0912345678	Current system rejects this value, but this is a defect against the stated requirement.
MFR04-EC-013	Phone number	Invalid	Contains letters.	09123abc78	Update is rejected.
MFR04-EC-014	Phone number	Invalid	Contains spaces or symbols.	091 234 5678	Update is rejected unless formatting is explicitly allowed.
MFR04-EC-015	Address	Valid	Normal shipping address.	123 Le Loi, Q1, TP HCM	Profile can be updated.
MFR04-EC-016	Address	Valid	Empty address if optional.	Empty string	Profile can be updated if address is optional.
MFR04-EC-017	Address	Valid	Empty or short address accepted by executed system.	Empty string / A	Profile can be updated when address is empty or short.
MFR04-EC-018	Email field	Valid	Email remains unchanged.	test@eshop.com	Email stays the same after update.
MFR04-EC-019	Email field	Invalid	User attempts to edit email through UI.	new@eshop.com	UI prevents edit or update ignores email change.
MFR04-EC-020	Profile ownership	Invalid	User attempts to update another user profile.	Other user ID	Update is rejected and other profile is unchanged.

EC ID	Input / Condition	Class Type	Description	Representative Value	Expected Result
MFR04-EC-021	Role attribute	Invalid	User attempts to change own role.	admin	Role remains unchanged.
MFR04-EC-022	Logout	Valid	User taps logout button.	Thoat	User is logged out.
MFR04-EC-023	Order history	Valid	User has one order.	Đơn #1	One order card is displayed.
MFR04-EC-024	Order history	Valid	User has no orders.	Empty order list	No-order message is displayed.
MFR04-EC-025	Order ownership	Invalid	Order history contains another user's order.	Other user's order	Other user's order is not displayed.
MFR04-EC-026	Order fields	Valid	Order has complete summary fields.	Date, total, status	Order number, date, total, and status are displayed.
MFR04-EC-027	Order amount	Valid	Positive total amount.	30.000.000 đ	Total amount is formatted correctly.
MFR04-EC-028	Order status	Valid	Pending order status.	Chờ xác nhận	Pending status is displayed and cancel action is available.
MFR04-EC-029	Cancel action	Valid	User cancels a pending order.	Tap Hủy đơn	Order is canceled or status/list updates according to confirmed rule.
MFR04-EC-030	Cancel action	Invalid	Cancel action on non-cancellable order.	Delivered/canceled order	Cancel action is hidden or rejected.

5. Domain Testing Test Cases

Test Case ID	Objective	Preconditions	Test Data	Steps	Expected Result	Actual Result	Status
MFR04-DT-001	Verify logged-in user can view profile page.	User is logged in.	Existing profile: test@eshop.com, Test User.	Open profile page.	Profile page displays email, full name, phone, address, update button, and logout button.	Profile page displays email, full name, phone, address, update button, and logout button.	Passed
MFR04-DT-002	Verify guest cannot access profile page.	User is logged out.	No session.	Open profile page URL/screen.	User is redirected to login	User is redirected to login	Passed
MFR04-DT-003	Verify valid profile update.	User is logged in.	Full name Test User Updated, phone 123456789, address 123 Le Loi, Q1.	Edit fields and tap Cập nhật.	Profile is saved and updated values remain after reload/reopen.	Profile is saved and update values remain after reload	Passed
MFR04-DT-004	Verify Vietnamese full name is accepted.	User is logged in.	Full name Nguyễn Hoàng Phi Hùng; other fields valid.	Edit full name and tap Cập nhật.	Profile is saved with the entered name.	Profile is saved with entered name	Passed
MFR04-DT-005	Verify empty full name is rejected.	User is logged in.	Full name empty; other fields valid.	Clear full name and tap Cập nhật.	Update is rejected and name validation is shown.	Profile is save with entered name	Failed
MFR04-DT-006	Verify whitespace-only full name is rejected.	User is logged in.	Full name " "; other fields valid.	Enter spaces in full name and tap Cập nhật.	Update is rejected after trimming and name validation is shown.	Profile is save with the whitespace-only full name	Failed
MFR04-DT-007	Verify 10-digit phone starting with 0 is accepted.	User is logged in.	Phone 0912345678; other fields valid.	Enter phone and tap Cập nhật.	Profile is saved.	Error message: "Số điện thoại không hợp lệ. Vui lòng nhập đúng 9-10 chữ số"	Failed

Test Case ID	Objective	Preconditions	Test Data	Steps	Expected Result	Actual Result	Status
MFR04-DT-008	Verify 11-digit phone starting with 0 is accepted.	User is logged in.	Phone 01234567890; other fields valid.	Enter phone and tap Cập nhật.	Update is rejected	Update is rejected	Passed
MFR04-DT-009	Verify 9-digit phone is accepted.	User is logged in.	Phone 091234567; other fields valid.	Enter phone and tap Cập nhật.	Profile is saved	Error message: "Số điện thoại không hợp lệ". Vui lòng đăng nhập 9-10 chữ số	Failed
MFR04-DT-010	Verify 11-digit phone is rejected.	User is logged in.	Phone 12345678901; other fields valid.	Enter phone and tap Cập nhật.	Update is rejected because phone length is greater than or equal to 11 digits.	Update is rejected because phone length is greater than or equal to 11 digits.	Passed
MFR04-DT-011	Verify phone not starting with 0 is rejected.	User is logged in.	Phone 123456789; other fields valid.	Enter phone and tap Cập nhật.	Update is rejected because phone must start with 0.	Profile is saved and update values remain after reload	Failed
MFR04-DT-012	Verify phone containing symbol is rejected.	User is logged in.	Phone 123*56789; other fields valid.	Enter phone and tap Cập nhật.	Update is rejected because phone must contain digits only.	Update is rejected because phone must contain digits only.	Passed
MFR04-DT-014	Verify default shipping address can be updated.	User is logged in.	Address 123 Le Loi, Q1, TP HCM; other fields valid.	Enter address and tap Cập nhật.	Address is saved as default shipping address.	Address is saved as default shipping address.	Passed
MFR04-DT-015	Verify empty address handling.	User is logged in.	Address empty; other fields valid.	Clear address and tap Cập nhật.	Profile is saved if address is optional; otherwise validation is shown.	Profile is saved if address is optional; otherwise validation is shown.	Passed
MFR04-DT-016	Verify email cannot be changed through UI.	User is logged in.	Current email test@eshop.com.	Attempt to focus/edit email field.	Email field is disabled/read-only and value cannot be changed through UI.	Email field is disabled/read-only and value cannot be changed through UI.	Passed
MFR04-DT-017	Verify update does not change email.	User is logged in.	Existing email test@eshop.com; valid editable fields.	Update full name/phone/address and reopen profile.	Email remains test@eshop.com.	Email remains test@eshop.com.	Passed
MFR04-DT-020	Verify logout from profile page.	User is logged in.	Tap Thoát.	Tap logout button.	User session ends and app shows logged-out state or login page.	User session ends and app shows logged-out state or login page	Passed
MFR04-DT-021	Verify order history section is displayed for a user with orders.	User is logged in and has at least one order.	Order #1 exists.	Open profile page and scroll to Lịch sử đơn hàng.	Order history section displays the user's order card.	Lịch sử đơn hàng displays Đơn #1.	Passed
MFR04-DT-022	Verify empty order history state.	User is logged in and has no orders.	Empty order list.	Open profile page and view Lịch sử đơn hàng.	Clear message such as Bạn chưa có đơn hàng nào. is displayed.	Clear message such as Bạn chưa có đơn hàng nào. is displayed.	Passed
MFR04-DT-024	Verify order summary fields are displayed.	User is logged in and has order #1.	Order date 2/7/2026, total 30.000.000 đ, status Chờ xác nhận.	Open profile page and view order card.	Order card displays order number, order date, total amount, and status.	Order card displays Đơn #1, Ngày đặt: 2/7/2026, Tổng tiền: 30.000.000 đ, and Trạng thái: Chờ xác nhận.	Passed
MFR04-DT-025	Verify pending order shows cancel button.	User is logged in and has a pending order.	Order status Chờ xác nhận.	Open profile page and view order card.	Hủy đơn button is shown for the pending order.	Hủy đơn button is shown.	Passed
MFR04-DT-026	Verify cancel pending order behavior.	User is logged in and has a cancellable pending order.	Tap Hủy đơn on order #1.	Tap cancel button and confirm if prompted.	Order is canceled or status/list updates according to confirmed product rule.	Order is canceled and status of order is updated "Đã hủy"	Passed

Test Case ID	Objective	Preconditions	Test Data	Steps	Expected Result	Actual Result	Status
MFR04-DT-027	Verify non-cancellable order does not show cancel button.	User is logged in and has a delivered/canceled order.	Status Đã giao or Đã hủy.	Open profile page and view order card.	Hủy đơn is hidden or cancel attempt is rejected.	Hủy đơn is hidden or cancel attempt is rejected.	Passed

6. Boundary Identification

Boundary ID	Field / Condition	Boundary Rule	Values to Test	Reason
MFR04-BD-001	Phone length lower boundary	Executed system behavior accepts 9 digits, although the stated requirement says 10-11 digits.	8, 9, 10 digits	Tests observed min - 1, observed min, and observed min + 1; records requirement mismatch.
MFR04-BD-002	Phone length upper boundary	Executed system behavior rejects 11 digits, although the stated requirement says 10-11 digits.	9, 10, 11 digits	Tests observed max - 1, observed max, and observed max + 1; records requirement mismatch.
MFR04-BD-003	Phone first character	Stated requirement says phone must start with 0, but execution shows inconsistent behavior.	First char 0, first char 1	Verifies leading-zero requirement and exposes mismatch.
MFR04-BD-004	Full name lower boundary	Full name is assumed required after trimming.	0, 1, 2 characters	Tests empty, minimum non-empty, and just above minimum.
MFR04-BD-006	Address length lower boundary	Address may be optional; requiredness must be confirmed.	0, 1, 2 characters	Verifies empty/short address behavior.
MFR04-BD-008	Order count boundary	Order history can contain zero, one, or multiple orders.	0, 1, 2+ orders	Verifies empty state, single order card, and repeated cards.
MFR04-BD-009	Order total lower boundary	Order total display should handle zero and positive values; negative-total behavior still needs controlled test data.	0, 1 VND	Verifies minimum display behavior from executed cases.
MFR04-BD-010	Order total display boundary	Large totals should remain readable and formatted.	999, 1.000, 30.000.000 VND	Verifies thousands separator and layout stability.

7. Boundary Value Analysis Test Cases

Test Case ID	Boundary Target	Preconditions	Test Data	Steps	Expected Result	Actual Result	Status
MFR04-BVA-001	Phone length below minimum	User is logged in.	Phone 12345678 (8 digits); other fields valid.	Submit profile update.	Update is rejected because phone must be 9-10 digits.	Update is rejected because phone must be 9-10 digits.	Passed
MFR04-BVA-002	Phone length at minimum	User is logged in.	Phone 123456789 (9 digits); other fields valid.	Submit profile update.	Profile is saved.	Profile is saved.	Passed
MFR04-BVA-004	Phone length above maximum	User is logged in.	Phone 12345678901 (11 digits); other fields valid.	Submit profile update.	Update is rejected because phone must be 9-10 digits.	Update is rejected because phone must be 9-10 digits.	Passed
MFR04-BVA-006	Phone leading zero valid boundary	User is logged in.	Phone 0912345678; other fields valid.	Submit profile update.	Profile is saved.	Error message: "Số điện thoại không hợp lệ. Vui lòng nhập đúng 9-10 chữ số"	Failed
MFR04-BVA-007	Phone leading zero invalid boundary	User is logged in.	Phone 123456789; other fields valid.	Submit profile update.	Update is rejected because phone does not start with 0.	Profile is saved	Failed
MFR04-BVA-008	Full name empty boundary	User is logged in.	Full name empty; other fields valid.	Submit profile update.	Update is rejected if full name is required.	Profile is saved	Passed
MFR04-BVA-009	Full name minimum non-empty boundary	User is logged in.	Full name A; other fields valid.	Submit profile update.	Profile is saved if one-character names are allowed.	Profile is saved	Passed
MFR04-BVA-010	Full name just above minimum boundary	User is logged in.	Full name An; other fields valid.	Submit profile update.	Profile is saved.	Profile is saved	Passed
MFR04-BVA-014	Address empty boundary	User is logged in.	Address empty; other fields valid.	Submit profile update.	Profile is saved if address is optional; otherwise validation is shown.	Profile is saved	Passed

Test Case ID	Boundary Target	Preconditions	Test Data	Steps	Expected Result	Actual Result	Status
MFR04-BVA-015	Address minimum non-empty boundary	User is logged in.	Address A; other fields valid.	Submit profile update.	Profile is saved if one-character address is allowed.	Profile is saved	Passed
MFR04-BVA-019	Order count zero boundary	User is logged in with no orders.	0 orders.	Open profile page.	Empty order-history message is displayed.	Empty order-history message is displayed.	Passed
MFR04-BVA-020	Order count one boundary	User is logged in with one order.	1 order.	Open profile page.	One order card is displayed without layout issues.	Don #1 is displayed.	Passed
MFR04-BVA-021	Order count multiple boundary	User is logged in with multiple orders.	2+ orders.	Open profile page.	Multiple order cards are displayed in order history.	Multiple order cards are displayed in order history.	Passed
MFR04-BVA-023	Order total zero boundary	User has an order with zero total.	Total 0 ₺.	Open profile page.	Zero total is displayed only if valid by product rule.	Zero total is displayed only if valid by product rule.	Passed
MFR04-BVA-024	Order total minimum positive boundary	User has an order with minimal positive total.	Total 1 ₺.	Open profile page.	Total is displayed with currency.	Zero total is displayed only if valid by product rule.	Passed
MFR04-BVA-025	Large order total display boundary	User has a high-value order.	Total 30.000.000 ₺.	Open profile page.	Total is formatted with thousands separators and does not break layout.	30.000.000 ₺ is displayed.	Passed